PSEUDO-COHERENT COMPLEXES ON A QUASI-COMPACT OPEN OF AN AFFINE SCHEME LIFT TO BOUNDED ABOVE FINITE FREE COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. The source flags '(Some details omitted; hint: clear denominators)' in the choice of beta; carried as an explicit obligation.

Editorial note. Inputs lemma-lift-quasi-coherent and lemma-affine-compare-bounded are same-chapter NOT carded (cross-line-claimed labels) and are declared external.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3377 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Pseudo-coherent objects are quasi-coherent. spaces-perfect.tex lemma-pseudo-coherent (carded cross-chapter b45-era as S9-perf-pseudo-coherent-qcoh-cohomology L26101; more general spaces version, same conclusion). E in D_QCoh(O_U).
- Lifting quasi-coherent objects over affines. lemma-lift-quasi-coherent (same chapter, NOT carded, declared external). E = E'|_U for some E' in D_QCoh(O_X).
- Affine comparison of D_QCoh with module complexes. lemma-affine-compare-bounded (same chapter, NOT carded, declared external). E' corresponds to a complex M-bullet of A-modules.
- Pseudo-coherence of localized module complexes. lemma-pseudo-coherent-affine (carded L26769). M-bullet_{f_j} is pseudo-coherent over A_{f_j}.
- Pseudo-coherence of cones and finiteness of cohomology. More on Algebra, lemmas cone-pseudo-coherent and finite-cohomology (cross-chapter, named). C-bullet_{f_j} is (n-1)-pseudo-coherent with finite H^{n-1}.

Source reference. Original reference 014E: focused-seeds/geometry/S9-perf2-lift-pseudo-coherent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08ED: sources/stacks-perfect/pinned-original-source.tex, source lines 3255–3255. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08E5: sources/stacks-perfect/pinned-original-source.tex, source lines 2422–2422. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 064R: focused-seeds/geometry/S9-perf2-lift-pseudo-coherent/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 064S: focused-seeds/geometry/S9-perf2-lift-pseudo-coherent/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be an affine scheme and let $U \subset X$ be a quasi-compact open subscheme. For any pseudo-coherent object E of $D(\mathcal{O}_U)$ there exists a bounded above complex of finite free $\mathcal{O}_X$-modules whose restriction to U is isomorphic to E.*

Proof. By Lemma 08E5 we see that E is an object of $D_{QCoh}(\mathcal{O}_U)$. By Lemma 08ED we may assume $E = E'|U$ for some object E' of $D_{QCoh}(\mathcal{O}_X)$. Write $X = \mathrm{Spec}(A)$. By Lemma 06Z0 we can find a complex $M^\bullet$ of A-modules whose associated complex of $\mathcal{O}_X$-modules is a representative of E'.

Choose $f_1, \ldots, f_r \in A$ such that $U = D(f_1) \cup \ldots \cup D(f_r)$. By Lemma 08E7 the complexes $M^\bullet_{f_j}$ are pseudo-coherent complexes of A_{f_j}-modules. Let n be an integer. Assume we have a map of complexes $\alpha : F^\bullet \to M^\bullet$ where $F^\bullet$ is bounded above, $F^i = 0$ for $i < n$, each F^i is a finite free R-module, such that

$$H^i(\alpha_{f_j}) : H^i(F^\bullet_{f_j}) \to H^i(M^\bullet_{f_j})$$

is an isomorphism for $i > n$ and surjective for $i = n$. Picture

$$\begin{array}{ccccccc} & & F^n & \longrightarrow & F^{n+1} & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \\ M^{n-1} & \longrightarrow & M^n & \longrightarrow & M^{n+1} & \longrightarrow & \ldots \end{array}$$

Since each $M^\bullet_{f_j}$ has vanishing cohomology in large degrees we can find such a map for $n \gg 0$. By induction on n we are going to extend this to a map of complexes

$F^\bullet \to M^\bullet$ such that $H^i(\alpha_{f_j})$ is an isomorphism for all i. The lemma will follow by taking $\widetilde{F^\bullet}$.

The induction step will be to extend the diagram above by adding F^{n-1}. Let $C^\bullet$ be the cone on α (Derived Categories, Definition 014E). The long exact sequence of cohomology shows that $H^i(C^\bullet_{f_j}) = 0$ for $i \geq n$. By More on Algebra, Lemma 064R we see that $C^\bullet_{f_j}$ is $(n-1)$-pseudo-coherent. By More on Algebra, Lemma 064S we see that $H^{n-1}(C^\bullet_{f_j})$ is a finite A_{f_j}-module. Choose a finite free A-module F^{n-1} and an A-module $\beta : F^{n-1} \to C^{n-1}$ such that the composition $F^{n-1} \to C^{n-1} \to C^n$ is zero and such that $F^{n-1}_{f_j}$ surjects onto $H^{n-1}(C^\bullet_{f_j})$. (Some details omitted; hint: clear denominators.) Since $C^{n-1} = M^{n-1} \oplus F^n$ we can write $\beta = (\alpha^{n-1}, -d^{n-1})$. The vanishing of the composition $F^{n-1} \to C^{n-1} \to C^n$ implies these maps fit into a morphism of complexes

$$\begin{array}{ccccccccc} & & F^{n-1} & \xrightarrow{d^{n-1}} & F^n & \longrightarrow & F^{n+1} & \longrightarrow & \ldots \\ & & \downarrow{\scriptstyle \alpha^{n-1}} & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \alpha} & & \\ \ldots & \longrightarrow & M^{n-1} & \longrightarrow & M^n & \longrightarrow & M^{n+1} & \longrightarrow & \ldots \end{array}$$

Moreover, these maps define a morphism of distinguished triangles

$$\begin{array}{ccccccc} (F^n \to \ldots) & \longrightarrow & (F^{n-1} \to \ldots) & \longrightarrow & F^{n-1} & \longrightarrow & (F^n \to \ldots)[1] \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle \beta} & & \downarrow \\ (F^n \to \ldots) & \longrightarrow & M^\bullet & \longrightarrow & C^\bullet & \longrightarrow & (F^n \to \ldots)[1] \end{array}$$

Hence our choice of β implies that the map of complexes $(F^{-1} \to \ldots) \to M^\bullet$ induces an isomorphism on cohomology localized at f_j in degrees $\geq n$ and a surjection in degree $n-1$. This finishes the proof of the lemma. $\square$